Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Main Topic: [Banana] – [Post-Harvest and Ratoon Crop.]

1.1 Subtopic 1: Ratoon Crop Lifecycle, Cropping Intensity, and Economics

1.1.1 1. Perennial Growth Dynamics and Generational Cycles

Banana (*Musa* spp.) exhibits a perennial growth habit wherein subterranean rhizomes continually produce successive generations of vegetative offshoots[1].

- **Generational Nomenclature:** The initial crop established from planted suckers or tissue-culture (TC) plantlets is termed the *plant crop* (mother plant)[12]. The vegetative sucker selected to succeed the harvested mother plant is the *follower* or *first ratoon crop*[13]. Subsequent cycles are designated as the second, third, and fourth ratoon crops[14].
- **Overlap and Lateral Bud Activation:** Lateral bud growth on the parent corm is released around flower bud initiation[3]. At the moment of inflorescence initiation, 11 un-emerged leaves remain inside the pseudostem; the growth of these 11 leaves plus the bunch filling period represents the overlapping growth phase between mother plant and ratoon follower[3].

1.1.2 2. Cropping Intensity and Multi-Cycle Production Schedules

By using uniform tissue-culture planting material and structured follower management, farmers can achieve high cropping intensity across multi-year cycles[4, 5].

- **Three-Crop 30-Month Calendar:** A standard production schedule yields 3 full crops—comprising 1 plant crop (12 months) and 2 sequential ratoon crops (9 months each)—within a total duration of 30 months without any yield reduction[5].
- **Accelerated 24–25 Month System:** In high-input production zones (such as Theni, Tamil Nadu), precise fertigation and temperature management allow up to 3 ratoon cycles to be completed within 24 to 25 months[4].
- **Irrigation Prerequisite:** Because the mother plant and developing ratoon follower simultaneously compete for soil water and mineral nutrients, multi-cycle ratooning requires an assured, year-round irrigation source[5].

1.1.3 3. Economic Advantages and Cost Structure

Ratooning significantly improves net farm returns by reducing capital expenditure and land preparation costs across subsequent crop cycles[5].

- **Input Cost Reduction:** Ratoon crop production eliminates major primary setup costs, including deep land ploughing, harrowing, pit excavation, and purchasing fresh tissue-culture plantlets[6].
- **Comparative Cost Breakdown:** Expenditure for the first ratoon crop is markedly lower than that of the plant crop (e.g., total cultivation cost drops from approximately $Rs.$ 95,310/$acre$ in the plant crop to approximately $Rs.$ 29,272/$acre$ in the first ratoon)[6].
- **Net Profitability:** Savings in land preparation, labor (reduced from 365 man-days to approximately 200 man-days), and planting material yield significantly higher net profit margins per unit area across a 2-year multi-crop cycle[6, 7].

1.2 Subtopic 2: Post-Harvest Stool Management & Mattacking Protocols

1.2.1 1. Mattacking Definition and Height Specifications

Mattacking is the cultural practice of cutting the mother plant pseudostem after bunch harvest to facilitate energy transfer to the succeeding ratoon generation[1].

- **Operational Execution:** Following bunch removal, the standing pseudostem is cut horizontally, leaving an intact stump of precisely **0.6 m (60 cm) height** above ground level[1, 3].
- **Prohibition of Ground-Level Cut:** Cutting the pseudostem flush at ground level immediately after harvest—a common practice among traditional growers—is strictly discouraged, as it arrests nutrient recycling and exposes the underground corm to premature rot[2].

1.2.2 2. Physiological Mechanisms of Carbohydrate and Nutrient Mobilization

The standing 0.6 m stump functions as a temporary storage and translocational organ for the developing follower[2, 3].

- **Sink-Source Energy Transfer:** Mobile carbohydrates, water, and residual mineral nutrients remaining in the pseudostem tissue continuously leach down into the connected corm[2, 3].
- **Daughter Follower Support:** These translocated reserves nourish the selected ratoon follower during its early establishment phase until the parent stump naturally withers, dries up, and detaches[2, 3].

1.2.3 3. Pathological and Entomological Modifications in Weevil-Endemic Fields

While mattacking is physiologically advantageous, its protocol must be modified in orchards under severe pest pressure[5, 6].

- **Pest Breeding Reservoir Risk:** In regions with high populations of Banana Pseudostem Borer (*Odoiporus longicollis*) or Rhizome Weevil (*Cosmopolites sordidus*), standing stumps provide ideal decaying tissue for adult egg laying and larval development[5, 6].
- **Ground-Level Removal and Chopping:** In weevil-endemic hotspots, standard mattacking is omitted[5, 6]. The harvested pseudostem is cut directly at the base, treated with an insecticide/biocontrol agent (e.g., 100 ml Carbaryl @ 2 g/L or *Beauveria bassiana*), and immediately chopped into small bits or shredded for soil mulching to destroy weevil harborages[5, 6].
- **Trap Conversion:** Alternatively, the felled pseudostem is cut into 30 cm longitudinal split bits and distributed across the orchard floor cut-side down to serve as monitor traps for adult weevil collection[6].

1.3 Subtopic 3: Ratoon Follower Selection, Desuckering, and Heading Back

1.3.1 1. Follower Selection Chronology and Selection Criteria

- **Vegetative Desuckering Rule:** Throughout the active vegetative phase of the mother (plant) crop, all emerging lateral suckers must be systematically removed to prevent competition for soil nutrients, moisture, and solar radiation [**?**, **?**, **?**, **?**].
- **Selection Post-Inflorescence Emergence:** A single follower is selected and retained **only after the inflorescence has emerged** (shooting phase) and all female hands have opened on the parent plant [**?**, **?**].
- **Sword Sucker Morphology:** The selected follower must be a vigorous *sword sucker* possessing a broad underground corm base, strong root attachment, and narrow, lanceolate blade leaves, which grow rapidly and shoot early compared to weak water suckers [**?**].

1.3.2 2. Systematic Desuckering and Offshoot Eradication Methods

Desuckering is the periodic removal of unwanted, non-selected lateral offshoots [**?**, **?**, **?**].

- **Operational Frequency:** Execute desuckering at regular monthly intervals (or every 5 to 6 weeks) from crop establishment up to harvest [**?**, **?**, **?**, **?**, **?**].
- **Mechanical Core Destruction:** Sever unwanted suckers cleanly above ground level, scoop out the central growing point (heart), or remove the sucker along with a small portion of corm tissue to delay re-sprouting [**?**, **?**, **?**].
- **Chemical Desuckering via Kerosene Core Pouring:** After cutting the unwanted sucker flush with the soil, pour **2 ml of kerosene oil** directly into the scooped central core to kill the apical meristem safely without causing systemic phytotoxicity to the parent stool [**?**, **?**, **?**].

1.3.3 3. Heading Back Technique for Ratoon Sucker Synchronization

- **Heading Back Execution:** Prune (head back) the selected ratoon sword sucker crosswise once at the **2-month stage** of its development [?].
- **Physiological Purpose:** Heading back temporarily arrests vertical apical dominance, forcing photosynthate allocation into the subterranean rhizome to enlarge corm diameter and build a robust root system before rapid vertical pseudostem elongation resumes [?].
- **Maturation Synchronization:** By the time the parent crop is harvested, the headed-back ratoon sucker is fully established, structurally balanced, and positioned for rapid, uniform shooting in the second cycle [?, ?].

1.4 Subtopic 4: Soil Health, Earthing Up, and Ratoon Nutrition

1.4.1 1. Nutrient Management and Fertigation Regimes for Ratoon Followers

To maintain yield stability across ratoon cycles, nutrient application must be fully replenished for the developing follower [?, ?, ?, ?].

- **Re-application of Recommended Dose of Fertilizers (RDF):** Apply 100% of the recommended fertilizer dose to the ratoon crop (200 g N, 60–70 g P_2O_5, and 300 g K_2O per plant) divided into split top-dressings at the 2nd, 4th, and 6th months post-harvest of the mother crop [?, ?, ?].
- **Organic and Bio-fertilizer Enrichment:** Basally incorporate 10 kg of well-decomposed Farm Yard Manure (FYM) and 1 kg Neem cake per mat around the follower root zone to encourage soil microbial activity and suppress soil-borne pathogens [?, ?, ?, ?].
- **Micro-Drip Fertigation Continuity:** Maintain daily drip fertigation schedules delivering water-soluble nitrogen (Urea @ 10–15 g/plant/week) and potassium (Muriate of Potash @ 8–10 g/plant/week) through peak vegetative growth up to shooting [?, ?, ?].

1.4.2 2. Earthing Up and Corm Coverage in Ratoon Systems

- **High-Matting Phenomenon:** In sequential ratoon crops, newly formed corms develop progressively higher on the parent rhizome base (high-matting effect), exposing upper root bases to desiccation and increasing wind lodging risks [?, ?, ?, ?].
- **Soil Mounding Protocol:** Mound loose topsoil 25 to 30 cm high around the pseudostem base of the follower sucker during early growth and monsoon land operations [?, ?, ?, ?].
- **Root Stimulation and Drainage Management:** Earthing up covers exposed corm tissues, anchors the expanding root system by stimulating new secondary cord roots, and converts inter-row lanes into deep drainage furrows to prevent waterlogging during rainy periods [?, ?, ?, ?].

1.4.3 3. Organic Mulching, Green Manuring, and Soil Moisture Conservation

- **Crop Residue and Leaf Trash Mulching:** Spread dry banana leaf trashes, shredded pseudostem bits, or sugarcane trash to a thickness of 10 to 15 cm over ratoon basins and inter-row channels [?, ?, ?, ?].
- **Soil Health Benefits:** Mulching conserves soil moisture, lowers soil surface temperatures during hot summer months, suppresses weed germination, and enriches soil organic carbon upon decomposition [?, ?, ?, ?].
- **In Situ Green Manuring:** Intercrop Sunnhemp (*Crotalaria juncea*) or Cowpea (*Vigna unguiculata*) in ratoon row alleys and incorporate biomass into the soil at 40 days post-sowing (flowering stage) to add organic nitrogen and improve soil structure [?, ?, ?, ?].

1.5 Subtopic 5: Integrated Pest & Disease Management in Ratoon Cycles

1.5.1 1. Corm/Rhizome Weevil and Nematode Management in Ratoon Mats

Ratoon crops are increasingly susceptible to soil-borne pest buildup if sanitation and bio-control measures are neglected across cycles [?, ?, ?, ?].

- **Split Pseudostem Trapping:** Deploy 100 split pseudostem log traps per hectare cut face down near ratoon mats [?, ?, ?, ?]. Inoculate cut faces with *Beauveria bassiana* (1×10^9 CFU/ml) or entomopathogenic nematodes (*Heterorhabditis indica*) to suppress adult *Cosmopolites sordidus* populations [?, ?, ?, ?].
- **Nematode Biological Control:** Burrowing nematodes (*Radopholus similis*) and root-knot nematodes (*Meloidogyne* spp.) cause root disorganization and mat uprooting in mature ratoon stools [?, ?, ?]. Apply Neem cake @ 500 g/plant along with bio-agents *Purpureocillium lilacinus* or *Trichoderma harzianum* @ 25 g/plant into ratoon root basins at six-month intervals [?, ?, ?].

1.5.2 2. Viral Vector Suppression and Chemical Mat Rogueing

Viral diseases can quickly spread through interconnected corm tissues in multi-cycle ratoon fields [?, ?, ?].

- **Routine Disease Scouting:** Inspect ratoon followers bi-weekly for viral diagnostic symptoms: "dot-dash" dark green vein streaking for Banana Bunchy Top Virus (BBTV), chlorotic mosaic for Cucumber Mosaic Virus (CMV), or reddish spindle streaks for Banana Bract Mosaic Virus (BBrMV) [?, ?, ?].
- **Herbicidal Eradication via 2,4-D Injection:** Instantly rogue virus-infected ratoon mats by injecting 5 to 10 ml of 2,4-D Sodium salt solution (25 g in 1 L water) into the pseudostem using a plant injector gun [?, ?, ?]. Complete wilting occurs within 8 to 12 days, killing the corm without allowing viral passage to adjacent stools [?, ?].

- **Systemic Vector Management:** Control banana aphid (*Pentalonia nigronervosa*) vector colonies by directed spraying of Imidacloprid 17.8 SL @ 0.2 ml/L or Dimethoate 30 EC @ 1.7 ml/L into leaf sheath axils [**?**, **?**, **?**].

1.5.3 3. Canopy Sanitation and Sigatoka Leaf Spot Control

- **De-trashing and Aeration:** Mechanically cut off and burn dry, hanging, or severely spotted lower leaves at monthly intervals to reduce ascospore inoculum pressure and increase canopy sunlight penetration [**?**, **?**, **?**].
- **Fungicidal Spray Rotation:** Alternate systemic sprays of Propiconazole 25 EC @ 1 ml/L or Carbendazim 50 WP @ 1 g/L with contact protectant Mancozeb 75 WP @ 2 g/L combined with 1% mineral oil adjuvant during monsoon flushes to prevent leaf spot buildup [**?**, **?**, **?**].